



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohammed Razia Alangir Banu
Student Name	N. Siri Chandana Reddy
Roll Number	2520030142
Branch	CSE
Course Outcome	CO5 (Counting Sort)

Case Study Topic: Waste Sense – Smart Waste Collection & Recycling Optimization System Using Counting

Work Done Summary:

The Waste Sense system uses Counting Sort to efficiently organize and analyze waste collection data. Counting Sort is particularly useful when sorting waste quantities, bin fill levels, or recycling statistics that fall within a known range of values. The algorithm provides fast sorting performance and helps generate reports for waste management operations.

Implementation Details:

1. Collect waste quantity data from smart bins.
2. Store the values in an array.
3. Determine the maximum waste quantity value.
4. Create a counting array to store the frequency of each value.
5. Count the occurrences of each waste quantity.
6. Reconstruct the sorted array using the counting array.
7. Display the sorted waste data for analysis.

Result: Screenshot/output:

```
Locations:
Collection Center
Bin A
Bin B
Bin C
Recycling Plant

BFS Traversal:
Collection Center -> Bin A -> Bin B -> Bin C -> Recycling Plant

DFS Traversal:
Collection Center -> Bin A -> Bin C -> Recycling Plant -> Bin B
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The Waste Sense system successfully applies Counting Sort to organize waste collection and recycling data. The algorithm efficiently sorts waste quantities and bin fill levels, enabling faster analysis, improved reporting, and better waste management planning.

GIT HUB LINK : <https://github.com/nemaliscreddy78-glitch/DSA2>

Student Signature	Faculty Signature